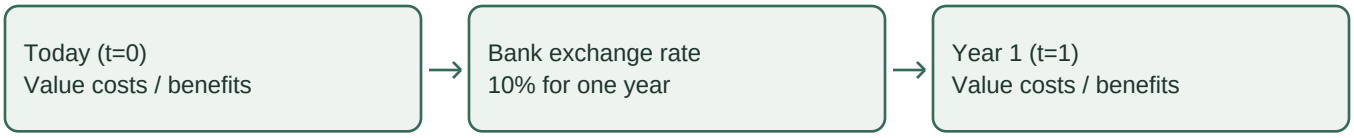
Market values, replication & NPV



Certain dollar cash flows. Borrowing and lending both available at 10% unless a panel states otherwise.

A. Same-day materials offer

Delivery	12 standardized units today
Competitive resale price	$50.00 per unit
Purchase cost today	$500.00

Purchase and sale are simultaneous. No transport costs, bid-ask spread, or quantity constraints. The buyer can resell every unit.

B. Bank and security quotes

Reference bank deposit today	$1,000.00
One-year bank rate	10%
Bond certain payoff at t=1	$1,100.00
Quoted bond price today	$980.00
Separate certain receipt at t=1	$1,210.00

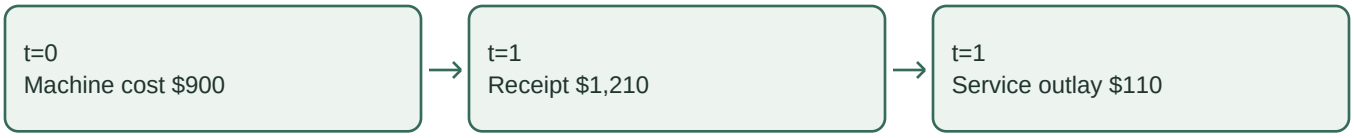
The bond and bank repayment have identical date, currency, and default risk. Borrowing is available to fund the bond purchase. Ignore trading costs and collateral restrictions.

C. Risk-state illustration

Claim	Weak economy	Strong economy
A	0	200
B	200	0

Each state is equally likely. A and B have the same average payout, but their payouts occur in different economic states. Do not assume their prices are equal.

Market values, replication & NPV



D. The receipt and service outlay happen on the same future date. All are certain.

D. Machine proposal

Initial equipment payment at t=0	$900.00
Receipt at t=1	$1,210.00
Required service payment at t=1	$110.00
Matched annual discount rate	10%
Other flows, taxes, transaction costs	None

Value every cost and benefit on one date. The service payment must be included even though it occurs after the purchase. Evaluate this project independently; it is not a choice between mutually exclusive projects.

E. Replication ledger - blank working copy

Position	Today	Year 1
Buy the bond from B	Pay quoted price	Receive $1,100
Borrow against its payoff	Receive loan principal	Repay $1,100
Combined position	Calculate net cash	Compare the two flows

Borrow exactly the present value of the future repayment. An arbitrage requires no negative future net payoff and a positive payoff somewhere, after costs. A merely optimistic forecast is not arbitrage.

F. Transaction cost variation

Buy one identical claim	Ask $101
Sell one identical claim	Bid $100
Observation	Last traded prices may differ from executable prices

Compare actual buying and selling prices, including fees. Without an executable profit, a displayed price discrepancy is not an arbitrage opportunity.